

Mathematics for Visual Computing

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August 26, 2025

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*Inspired by Math Essentials used for Visual Computing at the University of Otago

1 Introduction

This document provides the foundational mathematics essential for Visual Computing.

We will be dealing with three main types of mathematical objects: **scalars**, **vectors**, and **matrices**. Scalars are just ordinary numbers like 1, $-2\frac{3}{7}$, or π . Vectors are one-dimensional arrays of numbers, and matrices are two-dimensional arrays of numbers. It is important to distinguish between these types, because the operations you can perform on them and the results they produce are different. For example, while it makes sense to divide two scalars (as long as the divisor is not zero), division is not defined for vectors or matrices.

In these fields, vectors are used to represent geometric concepts like points, displacements, and directions. Matrices are used to represent transformations on these objects, such as scaling, translation, and rotation. By using vector and matrix notation, we gain access to a powerful algebra that allows us to express complex geometric operations.

2 Scalars

Scalar values can be represented on the real number line. A scalar may be interpreted either as a specific point on this line, or equivalently as a signed distance from the origin. In the latter interpretation, positive values correspond to displacements to the right and negative values to displacements to the left. Under this view, addition corresponds to the composition of displacements, while multiplication corresponds to the repeated application of a displacement.

2.1 Operations on Scalars

The standard arithmetic operations defined for scalars are addition, subtraction, multiplication, and division. Subtraction may be viewed as the addition of a negative number, while division is equivalent to multiplication by a reciprocal.

In addition, scalar operations satisfy several fundamental algebraic properties. These properties describe how operations behave when applied to more than one scalar:

- **Commutative Property.** An operation is *commutative* if the order of the operands does not affect the result. For example, when adding or multiplying scalars, it makes no difference whether a comes before b or vice versa:

$$a + b = b + a, \quad a \times b = b \times a \quad \forall a, b \in \mathbb{R}.$$

In contrast, subtraction and division are *not* commutative (e.g. $5 - 3 \neq 3 - 5$).

- **Associative Property.** An operation is *associative* if the way operands are grouped does not affect the result. For example, with addition and multiplication of scalars, it does not matter whether we combine the first two numbers or the last two:

$$a + (b + c) = (a + b) + c, \quad a \times (b \times c) = (a \times b) \times c \quad \forall a, b, c \in \mathbb{R}.$$

Again, subtraction and division are not associative (e.g. $(8 - 5) - 2 \neq 8 - (5 - 2)$).

- **Distributive Property.** Multiplication is said to *distribute* over addition. This means that multiplying a number by a sum gives the same result as multiplying it by each term separately and then adding the results:

$$a(b + c) = ab + ac \quad \forall a, b, c \in \mathbb{R}. \quad (1)$$

For example, $2(3 + 4) = 2 \times 3 + 2 \times 4 = 14$.

When both addition and multiplication appear in an expression, the usual *order of operations* applies: multiplication is performed before addition, unless brackets indicate otherwise.

2.2 A Note on Multiplication Symbols

There are many ways to write multiplication. The most usual is $a \times b$, but other common forms are a dot, $a \cdot b$, and just writing symbols side-by-side, ab . We will primarily use the latter form for scalars.

3 Trigonometric Functions

In the study of geometry and analysis, angles arise frequently, and their properties are described using trigonometric functions. In particular, we will make use of the three fundamental trigonometric functions: sine, cosine, and tangent. Angles may be expressed in different units of measure, and it is important to be familiar with both degrees and radians. While trigonometry is often introduced via right-angled triangles, an alternative and highly useful perspective is to consider their relationship to the unit circle.

3.1 Degrees and Radians

Although angles are commonly expressed in degrees, it is mathematically more natural to measure them in radians. A radian is defined as the length of the arc on the circumference of a circle of unit radius ($r = 1$) that subtends the given angle at the center. Since the circumference of the unit circle is 2π , a full revolution corresponds to 2π radians, which is equivalent to 360° . The conversion between degrees and radians is therefore given by:

$$s = \frac{\pi}{180^\circ} d, \quad d = \frac{180^\circ}{\pi} s. \quad (2)$$

3.2 Sine, Cosine, and Tangent

In terms of a right-angled triangle with hypotenuse H , opposite side O , and adjacent side A :

$$\sin \theta = \frac{O}{H}, \quad \cos \theta = \frac{A}{H}, \quad \tan \theta = \frac{O}{A}. \quad (3)$$

From Pythagoras' theorem, we get the fundamental identity:

$$\sin^2 \theta + \cos^2 \theta = 1. \quad (4)$$

3.3 Inverse Functions

The inverse functions are written as $\sin^{-1}(x)$, $\cos^{-1}(x)$, etc., which means 'the angle which has a sine of x '. They are also referred to as arcsine, arccosine, and arctangent (often 'asin', 'acos', 'atan' in programming).

4 Vectors

A vector is a one-dimensional array of numbers. We write vectors in bold face (e.g., $\mathbf{v}$) to distinguish them from scalars. By convention in this document, all vectors are column vectors.

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}. \quad (5)$$

The transpose operator, T , converts a column vector into a row vector: $\mathbf{v}^T = [v_1 \ v_2 \ \dots \ v_n]$.

4.1 Vector Addition and Subtraction

Vector addition and subtraction are defined by adding or subtracting the corresponding elements. This is only defined for vectors of the same size.

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}, \quad (6)$$

$$\mathbf{u} - \mathbf{v} = \begin{pmatrix} u_1 - v_1 \\ u_2 - v_2 \\ \vdots \\ u_n - v_n \end{pmatrix}. \quad (7)$$

4.2 Scalar Multiplication

Multiplying a vector by a scalar is performed element-wise. Given a scalar s and a vector $\mathbf{v}$, the product $\mathbf{u} = s\mathbf{v}$ has elements $u_i = s \cdot v_i$.

4.3 The Dot Product

The dot product (or scalar product) of two vectors of the same size, written $\mathbf{u} \cdot \mathbf{v}$, is a scalar value defined as:

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i. \quad (8)$$

The dot product is related to the angle θ between the vectors: $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$. Two non-zero vectors are perpendicular if and only if their dot product is 0.

4.4 Vector Lengths and Normalisation

The length (or magnitude, or norm) of a vector is written $||\mathbf{v}||$. It can be calculated from the dot product of the vector with itself.

$$||\mathbf{v}|| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}}. \quad (9)$$

A vector with length 1 is called a **unit vector**. The process of converting a vector to a unit vector by dividing it by its own length is called **normalisation**.

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{||\mathbf{v}||}. \quad (10)$$

4.5 Vector Cross Product

The cross product is defined only for 3-vectors and results in another 3-vector.

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}. \quad (11)$$

The resulting vector is perpendicular to both of the original vectors, unless they are parallel, in which case the cross product is the zero vector.

5 Matrices

A matrix is a two-dimensional array of numbers, written in uppercase letters (e.g., $\mathbf{A}$). Its size is expressed as [number of rows] $\times$ [number of columns]. The element in the i -th row and j -th column is denoted $a_{i,j}$.

$$\mathbf{A} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}. \quad (12)$$

5.1 Matrix Addition and Subtraction

Matrix addition and subtraction are defined element-wise, just as with vectors, and are only defined for matrices of the same size.

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{1,1} + b_{1,1} & \cdots & a_{1,n} + b_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} + b_{m,1} & \cdots & a_{m,n} + b_{m,n} \end{pmatrix} \quad (13)$$

$$(\mathbf{A} + \mathbf{B})_{i,j} = a_{i,j} + b_{i,j} \quad (14)$$

5.2 Scalar Multiplication

Multiplying a matrix by a scalar is also a per-element operation.

$$s\mathbf{A} = \begin{pmatrix} sa_{1,1} & \cdots & sa_{1,n} \\ \vdots & \ddots & \vdots \\ sa_{m,1} & \cdots & sa_{m,n} \end{pmatrix} \quad (15)$$

$$(sA)_{i,j} = s \cdot a_{i,j}. \quad (16)$$

5.3 Matrix Multiplication

The product of an $m \times p$ matrix $\mathbf{A}$ and a $p \times n$ matrix $\mathbf{B}$ is an $m \times n$ matrix $\mathbf{C}$. Note that the number of columns in $\mathbf{A}$ must match the number of rows in $\mathbf{B}$. The element $c_{i,j}$ is the dot product of the i -th row of $\mathbf{A}$ and the j -th column of $\mathbf{B}$.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \cdots & b_{pn} \end{pmatrix} \quad (17)$$

$$\mathbf{C} = \mathbf{AB} = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix} \quad (18)$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}, \quad 1 \leq i \leq m, \quad 1 \leq j \leq n. \quad (19)$$

$$\mathbf{C} = \begin{pmatrix} a_{11}b_{11} + \cdots + a_{1p}b_{p1} & a_{11}b_{12} + \cdots + a_{1p}b_{p2} & \cdots & a_{11}b_{1n} + \cdots + a_{1p}b_{pn} \\ a_{21}b_{11} + \cdots + a_{2p}b_{p1} & a_{21}b_{12} + \cdots + a_{2p}b_{p2} & \cdots & a_{21}b_{1n} + \cdots + a_{2p}b_{pn} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}b_{11} + \cdots + a_{mp}b_{p1} & a_{m1}b_{12} + \cdots + a_{mp}b_{p2} & \cdots & a_{m1}b_{1n} + \cdots + a_{mp}b_{pn} \end{pmatrix} \quad (20)$$

Matrix multiplication is associative, but it is ****not commutative**** ($AB \neq BA$ in general).

5.4 Matrix Transpose

The transpose of a matrix $\mathbf{A}$, denoted $\mathbf{A}^T$, is formed by swapping its rows and columns, so that $(\mathbf{A}^T)_{i,j} = A_{j,i}$.

$$\mathbf{A}^T = \begin{pmatrix} a_{1,1} & \cdots & a_{m,1} \\ \vdots & \ddots & \vdots \\ a_{1,n} & \cdots & a_{m,n} \end{pmatrix} \quad (21)$$

The transpose of a product of matrices reverses their order: $(AB)^T = B^T A^T$.

5.5 Matrices as Transforms of Vectors

We can think of a vector as an $n \times 1$ matrix. Multiplying an $m \times n$ matrix by an $n \times 1$ vector gives an $m \times 1$ vector as a result. This allows us to think of matrices as transformations that can rotate, scale, or shear points (represented by vectors) in space.

5.6 The Identity Matrix

The identity matrix, written $\mathbf{I}$, is a square matrix with 1s on the main diagonal and 0s elsewhere. It is the matrix equivalent of the number 1, as it represents a transform with no effect: $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$.

$$\mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (22)$$

5.7 Inverse Matrices

The inverse of a square matrix $\mathbf{M}$, denoted $\mathbf{M}^{-1}$, is a matrix such that $\mathbf{MM}^{-1} = \mathbf{M}^{-1}\mathbf{M} = \mathbf{I}$. Not all matrices have an inverse. A matrix that has an inverse is called invertible or nonsingular; one without is called singular. The inverse of a product reverses the order of multiplication: $(AB)^{-1} = B^{-1}A^{-1}$.

6 Appendix

6.1 Notation

$a, b, \dots$	Lower case letters in italics are scalars (simple numbers)
$\mathbf{a}, \mathbf{b}, \dots$	Lower case letters in bold are vectors (1-D arrays of numbers)
$A, B, \dots$	Upper case letters are matrices (2-D arrays of numbers)
$\sin^2(\theta)$	Means $(\sin(\theta))^2$, not $\sin(\sin(\theta))$. Similarly for cos and tan.
$\sin^{-1}(x)$	Arcsine, the angle whose sine is x . Similarly for cos and tan.

Matrices and Vectors

v_i	The i th element of a vector, $\mathbf{v}$
m_{ij}	The element of a matrix, M , in the i th row and j th column
$\mathbf{v}^T, M^T$	Transpose of a vector or matrix
$\mathbf{0}$	The zero-vector, $\mathbf{0} = [0 \ 0 \ \dots \ 0]^T$
$\mathbf{u} \cdot \mathbf{v}$	Dot product (a scalar value) of two vectors
$\mathbf{u} \times \mathbf{v}$	Cross product (a 3-vector) of two 3-vectors
$\ \mathbf{v}\ $	The length of a vector, $\ \mathbf{v}\ = \sqrt{\mathbf{v} \cdot \mathbf{v}}$
$\hat{\mathbf{v}}$	A unit vector in the same direction as $\mathbf{v}$, $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\ \mathbf{v}\ }$
$\mathbf{u} \perp \mathbf{v}$	The vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular to one another
$\mathbf{u} \parallel \mathbf{v}$	The vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel to one another
$\mathbf{I}$	The identity matrix, with diagonal entries of 1, and off-diagonal entries of 0
M^{-1}	The inverse of a matrix M . $MM^{-1} = I = M^{-1}M$

6.2 Quick reference

- Trigonometric functions of some common angles:

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
Degrees	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—

- $\sin^2(\theta) + \cos^2(\theta) = 1$.
- The angle between two vectors is

$$\cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right).$$

- Two non-zero vectors are perpendicular if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.
- Two unit vectors are parallel if and only if $\hat{\mathbf{u}} \cdot \hat{\mathbf{v}} = 1$.
- Two non-zero 3-vectors are parallel if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.